

SlabMaster REST API & SAP S/4HANA Integration

A modern, high-throughput RESTful JSON API designed for SAP enterprise resource planning, Project Systems (WBS), and field fabrication synchronization.

Why This API Supersedes Moraware

- **First-Class External ID Routing:** Use SAP WBS Elements, Purchase Orders, and Lot codes directly in REST paths and upsert payloads without internal UUID translation tables.
- **Full CRUD with Idempotent Upserts:** Eliminates duplicate creation errors caused by network re-transmissions.
- **Two-Way Change Data Capture (CDC):** SAP polls clean delta feeds to capture shop floor progress, digital templating signoffs, and completed installs.
- **Audit-Compliant Soft Cancellation:** Protects SAP cost accounting by flagging records as cancelled rather than breaking foreign keys.
- **Automated Dead-Letter Queue:** Outbound pushes to SAP use progressive backoff (1m ? 5m ? 15m ? 1h) with manual retry controls in the Admin Portal.

?? 2. Authentication & Security

Every REST request requires a tenant-scoped API key passed via standard HTTP headers:

```
# Recommended Header Format:
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45

# Or Standard Bearer Authorization:
Authorization: Bearer sm_live_9f83a1b4c7e28910fedcba45
```

Token Generation: Tenant administrators generate integration keys in the SlabMaster Admin Portal under **Settings ? API & ERP Integration**. Tokens are SHA-256 hashed prior to storage and can be scoped with **read**, **write**, **sync**, and **delete** permissions.

??? 3. SAP ERP Data Field Mapping Matrix

This table maps standard SAP S/4HANA & ECC entities to SlabMaster's 3-tier hierarchy:

SAP Entity / Table	SAP Field Name	SlabMaster Route / Field	Description & Role
Customer Master (KNA1)	KUNNR / NAME1	Account.external_id	Builder / General Contractor Master
Real Estate / Subdivision	SWERK / OBJNR	Community.external_id	Residential Development / Subdivision
Unit / Building Plot	PARKZ / Unit ID	Lot.external_id	Physical Lot / Unit (1 Lot ? Many Jobs)
Sales Order (VBAK)	VBELN	Job.external_id	Fabrication & Install Work Order
WBS Element (PRPS)	POSID / PSPNR	Job.project_number	Cost Collector & Project Milestone
Plant / Facility (T001W)	WERKS	Region.region_code	Regional Fabrication Shop (e.g. ATL, PHX)
Milestone / Phase	MLST / AUFNR	JobActivity.external_id	Phase (Template, Fab, Install)

3.1 Live Demo Test Fixtures (Moraware Crawled Data & SAP External IDs)

The test subscriber environment is seeded with actual production hierarchy records crawled from Moraware, mapped directly to enterprise SAP S/4HANA external identifiers across all five tiers:

Entity Tier	Display Name in Moraware	SAP External ERP ID	Parent Reference
Account	LENNAR HOMES OF GEORGIA - 126954	SAP-CUST-126954	Top-Level Tenant
Community	AUSTIN PARK AT HIGHLAND (LNXAUS)	SAP-COMM-LNXAUS	account_id: SAP-CUST-126954
Lot	000036 (5436 Austin Park Way)	SAP-LOT-LNXAUS-036	community_id: SAP-COMM-LNXAUS
Job (Order)	LNXAUS_000036_000_01 (SO-10170)	SAP-SO-10170	lot_id: SAP-LOT-LNXAUS-036
Activities (WBS)	Template, CAD, Sawjet, Polish, Install, Sign-off	SAP-WBS-10170-TMPL ... - SIGN	job_id: SAP-SO-10170
Account	CENTURY COMMUNITIES SOUTHEAST - 122973	SAP-CUST-122973	Top-Level Tenant
Community	OLD GREENWOOD GLEN (CS2OGG)	SAP-COMM-CS2OGG	account_id: SAP-CUST-122973
Lot	000069 (1869 Greenwood Ridge)	SAP-LOT-CS2OGG-069	community_id: SAP-COMM-CS2OGG
Job (Order)	CS2OGG_000069_000_01 (SO-10720)	SAP-SO-10720	lot_id: SAP-LOT-CS2OGG-069
Account	CENTURY COMMUNITIES NASHVILLE - 125316	SAP-CUST-125316	Top-Level Tenant
Community	AUTUMN LAKES NASHVILLE (CNAALN)	SAP-COMM-CNAALN	account_id: SAP-CUST-125316
Job (Order)	CNAALN_000033_000_01 (SO-10485)	SAP-SO-10485	lot_id: SAP-LOT-CNAALN-033
Account	DREAM FINDERS HOMES NASHVILLE - 130470	SAP-CUST-130470	Top-Level Tenant
Community	DREAM CREST (DF41DC)	SAP-COMM-DF41DC	account_id: SAP-CUST-130470
Job (Order)	DF41DC_0000DC_000_01 (SO-10941)	SAP-SO-10941	lot_id: SAP-LOT-DF41DC-001
Account	PERRY HOMES OF FLORIDA LLC - 129495	SAP-CUST-129495	Top-Level Tenant
Community	STAR FARMS LWR 90'S	SAP-COMM-STARFARMS	account_id: SAP-CUST-129495
Job (Warranty)	P2HSPN_001079_000_01 (Warranty Chip Repair)	SAP-SO-1079	activity: SAP-WBS-1079-CHIP



Postman Demo Tip: All fixtures above are included in *Folder 7: Moraware Live Demo Fixtures* in the official Postman collection for 1-click execution.

4. REST CRUD Endpoints

All endpoints support idempotent upserts, external ID resolution, standard HTTP status codes, and JSON payloads. All mutations automatically publish to the Two-Way CDC stream.

4.1 Builder Accounts (Top-Level Hierarchy)

Represents the top-level builder or parent contractor entity (e.g. Lennar, Toll Brothers, Pulte).

POST /api/v1/accounts/upsert

IDEMPOTENT UPSERT

```
POST /api/v1/accounts/upsert HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
Content-Type: application/json

{
  "external_id": "SAP-BLD-01",
  "name": "Lennar Homes - Central Division",
  "account_type": "Builder",
  "billing_address": "1200 Builder Way, Suite 400",
  "city": "Austin",
  "state": "TX",
  "zip_code": "78701",
  "phone": "512-555-0199",
  "payment_terms": "NET30"
}
```

Response (HTTP 200 OK / 201 Created):

```
{
  "message": "Builder Account synchronized successfully.",
  "operation": "UPsertED",
  "data": {
    "accountUuid": "d3b07384-d113-4944-9c02-76342cbe8a01",
    "externalId": "SAP-BLD-01",
    "name": "Lennar Homes - Central Division",
    "status": "Active",
    "syncedAt": "2026-09-04T19:30:00.000Z"
  }
}
```

GET /api/v1/accounts/by-external-id/{externalId}

LOOKUP

```
GET /api/v1/accounts/by-external-id/SAP-BLD-01 HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

DELETE**/api/v1/accounts/by-external-id/{externalId}****SOFT DEACTIVATE**

Flags the account as inactive while preserving historical work orders, cut tickets, and slab assignments.

```
DELETE /api/v1/accounts/by-external-id/SAP-BLD-01 HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

```
{
  "message": "Account marked inactive.",
  "operation": "DEACTIVATED",
  "externalId": "SAP-BLD-01"
}
```

4.2 Communities (Subdivisions / Developments)

Represents a development neighborhood or subdivision nested under a Builder Account.

POST /api/v1/communities/upsert

IDEMPOTENT UPSERT

```
POST /api/v1/communities/upsert HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
Content-Type: application/json

{
  "external_id": "COMM-HIGHLAND",
  "account_external_id": "SAP-BLD-01",
  "name": "Highland Ranch Estates",
  "region": "Southwest",
  "city": "Dripping Springs",
  "state": "TX",
  "zip_code": "78620",
  "contact_name": "Marcus Brody",
  "contact_phone": "512-555-0244"
}
```

Response (HTTP 200 OK):

```
{
  "message": "Community synchronized successfully.",
  "operation": "UPSERED",
  "data": {
    "communityUuid": "719d3810-6c92-4f35-901a-829d71c89012",
    "externalId": "COMM-HIGHLAND",
    "accountExternalId": "SAP-BLD-01",
    "name": "Highland Ranch Estates"
  }
}
```

GET /api/v1/communities/by-external-id/{externalId}

DELETE /API/V1/COMMUNITIES/BY-EXTERNAL-ID/{EXTERNALID}

```
GET /api/v1/communities/by-external-id/COMM-HIGHLAND HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```


4.3 Lots (Job Sites / Addresses)

Represents a specific building parcel or home site linked to a Builder Community.

POST /api/v1/lots/upsert

IDEMPOTENT UPSERT

```
POST /api/v1/lots/upsert HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
Content-Type: application/json

{
  "external_id": "LOT-HIGHLAND-14",
  "account_external_id": "SAP-BLD-01",
  "community_external_id": "COMM-HIGHLAND",
  "lot_number": "14",
  "block_number": "C",
  "street_address": "10428 E Sunburst Trail",
  "target_install_date": "2026-09-22"
}
```

Response (HTTP 200 OK):

```
{
  "message": "Lot synchronized successfully.",
  "operation": "UPsertED",
  "data": {
    "lotUuid": "b4a8e910-3810-410a-81a2-918274019283",
    "externalId": "LOT-HIGHLAND-14",
    "lotNumber": "14",
    "communityExternalId": "COMM-HIGHLAND"
  }
}
```

GET /api/v1/lots/by-external-id/{externalId}

DELETE /API/V1/LOTS/BY-EXTERNAL-ID/{EXTERNALID}

```
GET /api/v1/lots/by-external-id/LOT-HIGHLAND-14 HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

4.4 Jobs (Fabrication & Installation Work Orders)

The core production unit in SlabMaster containing slab allocations, cut lists, and scheduling.

POST /api/v1/jobs/upsert

IDEMPOTENT UPSERT

```
POST /api/v1/jobs/upsert HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
Content-Type: application/json

{
  "external_id": "SAP-ORD-90210",
  "account_external_id": "SAP-BLD-01",
  "community_external_id": "COMM-HIGHLAND",
  "lot_external_id": "LOT-HIGHLAND-14",
  "job_name": "Kitchen Perimeter & Waterfall Island",
  "job_category": "INITIAL_INSTALL",
  "project_number": "PRJ-2026-904",
  "sales_order_number": "SO-881920",
  "target_install_date": "2026-09-22T00:00:00Z"
}
```

Response (HTTP 200 OK):

```
{
  "message": "Job updated successfully.",
  "operation": "UPDATED",
  "data": {
    "jobUuid": "a823e59b-12d4-42ea-8910-cba190827364",
    "externalId": "SAP-ORD-90210",
    "jobName": "Kitchen Perimeter & Waterfall Island",
    "status": "Active"
  }
}
```

GET /api/v1/jobs/by-external-id/{externalId}

DELETE /API/V1/JOBS/BY-EXTERNAL-ID/{EXTERNALID}

```
GET /api/v1/jobs/by-external-id/SAP-ORD-90210 HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

4.5 Job Activities (Production & Field Milestones)

Detailed tracking of lifecycle steps: Laser Template, CAD Programming, CNC Sawjet Cutting, Edge Polishing, Field Installation, and Warranty Sign-off.

POST /api/v1/activities/upsert

IDEMPOTENT UPSERT

```
POST /api/v1/activities/upsert HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
Content-Type: application/json

{
  "external_id": "ACT-90210-INSTALL",
  "job_external_id": "SAP-ORD-90210",
  "activity_type": "INSTALLATION",
  "name": "Field Countertop Installation & Seaming",
  "status": "Scheduled",
  "scheduled_date": "2026-09-22T08:00:00Z",
  "assigned_crew": "Crew-Alpha-Austin",
  "estimated_hours": 6.5
}
```

Response (HTTP 200 OK / 201 Created):

```
{
  "message": "Activity synchronized successfully.",
  "operation": "UPsertED",
  "data": {
    "activityUuid": "f9082341-a1b2-4c3d-8e9f-012938475610",
    "externalId": "ACT-90210-INSTALL",
    "jobExternalId": "SAP-ORD-90210",
    "activityType": "INSTALLATION",
    "status": "Scheduled",
    "scheduledDate": "2026-09-22T08:00:00Z"
  }
}
```

GET /api/v1/activities/by-external-id/{externalId}

DELETE /API/V1/ACTIVITIES/BY-EXTERNAL-ID/{EXTERNALID}

```
GET /api/v1/activities/by-external-id/ACT-90210-INSTALL HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

?? 5. SAP ABAP Integration Code Sample

SAP developers can use the standard `cl_http_client` class to push orders from SAP S/4HANA or ECC to SlabMaster:

```

REPORT z_slabmaster_job_upsert.

DATA: lo_http_client TYPE REF TO if_http_client,
      lv_url          TYPE string VALUE 'https://dev.slabmasterapp.com/api/v1/jobs/upsert',
      lv_api_key      TYPE string VALUE 'sm_live_9f83a1b4c7e28910fedcba45',
      lv_json         TYPE string,
      lv_response     TYPE string,
      lv_subrc        TYPE sy-subrc.

" Build JSON payload from SAP Sales Order / WBS
lv_json = '{"external_id":"SAP-ORD-90210",' &&
         '"account_external_id":"SAP-BLD-01",' &&
         '"community_external_id":"COMM-HIGHLAND",' &&
         '"lot_external_id":"LOT-HIGHLAND-14",' &&
         '"job_name":"Kitchen Perimeter & Island",' &&
         '"target_install_date":"2026-09-22T00:00:00Z"}'.

" Create HTTP Client by URL
cl_http_client=>create_by_url(
  EXPORTING url      = lv_url
  IMPORTING client = lo_http_client ).

" Set Headers
lo_http_client->request->set_method( if_http_request=>co_request_method_post ).
lo_http_client->request->set_header_field( name = 'X-API-Key' value = lv_api_key ).
lo_http_client->request->set_header_field( name = 'Content-Type' value = 'application/json' ).
lo_http_client->request->set_cdata( lv_json ).

" Send and Receive
lo_http_client->send( ).
lo_http_client->receive(
  EXCEPTIONS
    http_communication_failure = 1
    http_invalid_state        = 2
    http_processing_failed    = 3
    OTHERS                    = 4 ).

IF sy-subrc = 0.
  lv_response = lo_http_client->response->get_cdata( ).
  WRITE: / 'SlabMaster Synchronized Successfully:', lv_response.
ELSE.
  WRITE: / 'Error Calling SlabMaster API:', sy-subrc.
ENDIF.

lo_http_client->close( ).

```

?? 6. Two-Way Change Data Capture (CDC)

To capture shop floor progress and installation signoffs without firewall-blocked webhooks, SAP middleware polls the delta feed:

```
GET /api/v1/sync/changes?since=2026-09-04T12:00:00Z&limit=100 HTTP/1.1
Host: dev.slabmasterapp.com
X-API-Key: sm_live_9f83a1b4c7e28910fedcba45
```

Delta Event Response:

```
{
  "data": [
    {
      "id": 1042,
      "entityType": "JOB",
      "entityId": "a823e59b-12d4-42ea-...",
      "changedByName": "Field Lead Tablet (PWA)",
      "changeSummary": "Install sign-off completed with customer signature",
      "fieldName": "status",
      "oldValue": "In Fabrication",
      "newValue": "Installed & Signed",
      "createdAt": "2026-09-04T18:45:12.000Z"
    }
  ],
  "metadata": {
    "count": 1,
    "since": "2026-09-04T12:00:00Z",
    "cursor": "2026-09-04T18:45:12.000Z",
    "hasMore": false
  }
}
```

7. Outbound Resilience & Dead-Letter Queue

When SlabMaster pushes milestones back to SAP and encounters network interruptions or SAP maintenance downtime, events enter the **Outbound ERP Retry Queue** with automated progressive backoff:

Attempt #	Backoff Delay	Target Failure Type
Attempt 1	Immediate (0s)	Transient socket/TLS negotiation resets
Attempt 2	1 Minute	Brief microservice restarts

Attempt #	Backoff Delay	Target Failure Type
Attempt 3	5 Minutes	Rate limit (HTTP 429) backoff
Attempt 4	15 Minutes	Load balancer failovers
Attempt 5	1 Hour	Scheduled maintenance windows
Exhausted	Stuck / Dead-Letter (FAILED)	Requires admin review & manual retry trigger

Manual Recovery Endpoints

```
# Retry a specific stuck item:
POST /api/v1/sync/queue/queue-tx-9941/retry

# Batch retry all failed items in the dead-letter queue:
POST /api/v1/sync/queue/retry-all
```

?? 8. Testing with Postman

You can test every endpoint immediately using our official pre-configured Postman Collection:

- 1. Download the collection: [slabmaster_postman_collection.json](#)
- 2. Open Postman ? Click **Import** ? Select the JSON file.
- 3. Navigate to the collection's **Variables** tab and enter your active API token in the `apiKey` field.
- 4. Execute queries against `{{baseUrl}}/api/v1/...` with automatic authorization!